1. 实际上 上一次作业已证明过.

$$P\left(\bigcup_{i=1}^{n} E_i\right) = P(E_1) + \sum_{i=2}^{n} P(E_1^c \cdots E_{i-1}^c E_i)$$

而由德摩根定律

$$\left(\bigcap_{k=1}^{i-1} E_k^c\right)^c = \left[\left(\bigcup_{k=1}^{i-1} E_k\right)^c\right]^c = \bigcup_{k=1}^{i-1} E_k$$

$$\Rightarrow \quad P\left(\left(\bigcap_{k=1}^{i-1} E_k^c\right) \cap E_i\right) = P(E_i) - P\left(\bigcup_{k=1}^{i-1} E_i E_k\right)$$

$$\Rightarrow \quad P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i) - \sum_{i=2}^{n} P\left(\bigcup_{k=1}^{i-1} E_i E_k\right).$$

那么 $P(A+B+C)$

$$= P(A)+P(B)+P(C) - P(AB) - P(AC+BC)$$

$$= P(A)+P(B)+P(C) - P(AB) - P(AC) - P(BC) + P(ABC).$$

z.

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ps：本来想直接导出容斥原理的，但卡住了.

另外. 考虑集合的示性函数，容易发现：

$$1_{\bigcup_{i=1}^{n} E_i} = 1 - \prod_{i=1}^{n} (1 - 1_{E_i})$$

$$= \sum_{r=1}^{n} (-1)^{r+1} \sum_{i_1 < \cdots < i_r} 1_{\bigcap_{k=1}^{r} E_{i_k}}$$ （这里应用了 $1_{E_i \cap E_j} = 1_{E_i} \cdot 1_{E_j}$）

再对两端取期望. 有

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{r=1}^{n} (-1)^{r+1} \sum_{i_1 < \cdots < i_r} P\left(\bigcap_{k=1}^{r} E_{i_k}\right).$$

代入 $n=3$ 即证.

T_2.

$\forall A \subset \Omega$.

1° $P(A|B) = \dfrac{P(AB)}{P(B)} \geq 0$

2° $P(\Omega | B) = \dfrac{P(\Omega B)}{P(B)} = \dfrac{P(B)}{P(B)} = 1$

3° $\forall i \neq j$ $A_i A_j = \phi$

$$\Rightarrow P\left(\bigcup_{i=1}^{\infty} A_i \,\Big|\, B\right) = \frac{P\left(B \bigcup_{i=1}^{\infty} A_i\right)}{P(B)}$$

$$= \frac{P\left(\bigcup_{i=1}^{\infty} A_i B\right)}{P(B)}$$

$$= \frac{\sum_{i=1}^{\infty} P(A_i B)}{P(B)}$$

$$= \sum_{i=1}^{\infty} P(A_i | B).$$

$\blacksquare$.

T_3

<1> F 若 $A \subsetneq B \subsetneq \Omega$

$$\Rightarrow P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} > P(A)$$

<2> F ϕ、Ω.

<3> F 一个很常见的例子.

掷两次色子.

$A := \{$第1次为 $1, 2, 3\}$

$B := \{$第1次为 $3, 4, 5\}$

$C := \{$两次和为$9\}$

$$\Rightarrow P(A) = P(B) = \frac{1}{2} \qquad P(C) = \frac{1}{9}$$

$$P(ABC) = \frac{1}{36} = P(A)P(B)P(C)$$

而 $P(AB) = \frac{1}{6} \neq \frac{1}{4} = P(A)P(B)$.

T_4

A_2、A_3 独立.

A_2, A_5 不独立.

易见 $P(A_2) = \frac{1}{2} \qquad P(A_3) = \frac{1}{3} \qquad P(A_5) = \frac{7}{36}$

$$P(A_2A_3) = \frac{1}{6} \qquad P(A_2A_5) = \frac{1}{12}.$$

T_5

独立 $\nRightarrow$ 条件独立

$\Omega = \{1, 2, 3, 4\}$ $\quad P(A) = P(B) = \frac{1}{2}$

$A = \{1, 2\}$ $\quad P(AB) = \frac{1}{4} = P(AB)$

$B = \{2, 3\}$ $\quad$ 而 $P(A|C) = P(B|C) = \frac{2}{3}$

$C = \{1, 2, 3\}$ $\quad P(AB|C) = \frac{1}{3} \neq P(A|C)P(B|C)$

条件独立 $\nRightarrow$ 独立.

$\Omega = \{1, 2, 3\}$ $\quad P(A|C) = P(B|C) = 0$

$A = \{1\}$ $\quad P(AB|C) = 0$

$B = \{2\}$ $\quad$ 而 $P(A) = P(B) = \frac{1}{3}$

$C = \{3\}$ $\quad P(AB) = 0$.

T_6

重复 n 次都不发生的概率为 $(1-\varepsilon)^n$

而 $\lim_{n\to\infty} 1-(1-\varepsilon)^n = 1$

□.

T_7

$A := \{$朝上的一面为红$\}$

$B := \{$朝下的一面为黑$\}$.

$$\Rightarrow P(B|A) = \frac{P(BA)}{P(A)} = \frac{\frac{1}{3}\times\frac{1}{2}}{\frac{1}{3}+\frac{1}{3}\times\frac{1}{2}} = \frac{1}{3}.$$

T_8 <1> 显然公平.

<2> $A_i := \{$第 i 个人抓中 "中"$\}$

$$\Rightarrow P(A_i) = P\left(A_i\left(\bigcap_{j=1}^{i-1} A_j^c\right)\right)$$

$$= P(A_1^c)\cdot P(A_2^c|A_1^c)\cdot P(A_3^c|A_1^cA_2^c)\cdots P\left(A_i\,\middle|\,\bigcap_{j=1}^{i-1}A_j^c\right)$$

$$= \frac{n-1}{n}\cdot\frac{n-2}{n-1}\cdots\frac{n-i+1}{n-i+2}\cdot\frac{1}{n-i+1} = \frac{1}{n}.$$

即公平.

T_9.

$A := \{$小明患病$\}$

$B := \{$检验结果为阳性$\}$

$P(A) = 0.6$

$P(B|A^c) = 0.3$ $\qquad P(B|A) = \frac{P(BA)}{P(A)} = 1 \Rightarrow P(BA) = 0.6$

$$P(B) = P(B|A^c)\cdot P(A^c) + P(B|A)\cdot P(A) = 0.72.$$

$$\Rightarrow P(A|B) = \frac{P(AB)}{P(B)} = \frac{0.6}{0.72} \approx 0.833 > 0.8$$

$\therefore$ 仍应建议手术.

T_{10} <1> 记刚开始有 i 个游戏币时赢取到 n 个的概率为 A_i.

则 $A_0=0 \quad A_n=1$

$$A_i = pA_{i-1} + (1-p)A_{i+1}$$

$$\Rightarrow \quad A_{i+1}-A_i = \frac{p}{1-p}(A_i - A_{i-1})$$

$$\Rightarrow \quad A_i - A_1 = \sum_{k=1}^{i-1}\left(\frac{p}{1-p}\right)^k \cdot A_1$$

$$= \left[\frac{1-\left(\frac{p}{1-p}\right)^i}{1-\frac{p}{1-p}} - 1\right]A_1 \quad (p \neq 0.5)$$

令 $i=N$

$$\Rightarrow \quad A_1 = \frac{1-\frac{p}{1-p}}{1-\left(\frac{p}{1-p}\right)^i} \quad \Rightarrow \quad A_i = \frac{1-\left(\frac{p}{1-p}\right)^i}{1-\left(\frac{p}{1-p}\right)^n}$$

又最后一定会赢钱离场或输光离场

那么输光离场的概率为 $P_i = 1 - \frac{1-\left(\frac{p}{1-p}\right)^i}{1-\left(\frac{p}{1-p}\right)^n}$

$p=0.5$ 时. 有 $\quad A_i - A_1 = (i-1)A_1$

$$\Rightarrow \quad A_1 = \frac{1}{n} \quad \Rightarrow \quad A_i = \frac{i}{n}. \quad \Rightarrow \quad P_i = 1-\frac{i}{n}.$$

<2> $p=0.5$ 时 $\quad P_i = 1-\frac{i}{n} \to 1$

$p<0.5$ 时 $\quad \frac{p}{1-p} < 1 \qquad \left(\frac{p}{1-p}\right)^n \to 0$

$$P_i \to 1$$

即输光离场的概率极限为 1.

T_{11}

$A := \{\text{生物最终灭亡}\}$

$B_1 := \{\text{生物一代后灭亡}\}$

$B_2 := \{\text{生物一代后不变}\}$

$B_3 := \{\text{生物一代后}\times 2\}$.

由全概率公式.

$$P(A) = P(A|B_1)\cdot P(B_1) + P(A|B_2)\cdot P(B_2) + P(A|B_3)\cdot P(B_3)$$

$$= 1\cdot\frac{1}{3} + P(A)\cdot\frac{1}{3} + P^2(A)\cdot\frac{1}{3}$$

$$\Rightarrow P(A) = 1.$$

即时间足够长后一定会灭亡。!!

T_{12}

$$P_{甲} = 0.8\times 0.8 + 0.1\times 0.05 + 0.1\times 0.1$$

$$= 0.655$$

$$P_{乙} = 0.6\times 0.8 + 0.9\times 0.05 + 0.9\times 0.1$$

$$= 0.665$$

$\Rightarrow$ 乙

甲？

T_{13}

<1> $P(B_1)=\frac{1}{2}\times\frac{4}{5}+\frac{1}{2}\times\frac{2}{5}=\frac{3}{5}$

$$P(U_1|B_1)=\frac{P(U_1B_1)}{P(B_1)}=\frac{\frac{1}{2}\times\frac{4}{5}}{\frac{3}{5}}=\frac{2}{3}$$

$P(U_1)=\frac{1}{2}<P(U_1|B_1)$

U_1 黑球占比比较多 所以摸到黑球的话

更可能是摸的 U_1

<2> 放回：$P(B_2)=\frac{1}{2}\times\frac{4}{5}+\frac{1}{2}\times\frac{2}{5}=\frac{3}{5}$

不放回：$P(B_2)=\frac{1}{2}\times\left(\frac{4}{5}\times\frac{3}{4}+\frac{1}{5}\times1\right)+\frac{1}{2}\times\left(\frac{2}{5}\times\frac{1}{4}+\frac{3}{5}\times\frac{1}{2}\right)$

$=\frac{3}{5}$

相等！

这是由于并未引入新的观测信息。

<3> $P(B_2|B_1)=\dfrac{P(B_2B_1)}{P(B_1)}$

$$=\frac{\frac{1}{2}\times\frac{4}{5}\times\frac{4}{5}+\frac{1}{2}\times\frac{2}{5}\times\frac{2}{5}}{\frac{3}{5}}$$

$=\frac{2}{3}>\frac{3}{5}$

这是由于第一次取出黑球增大了 U_1 的可能性。

<4>

$$P(B_{n+1} \mid B_1B_2\cdots B_n)$$

$$= \frac{P(B_1B_2\cdots B_{n+1})}{P(B_1B_2\cdots B_n)}$$

$$= \frac{\frac{1}{2}\cdot\left(\frac{4}{5}\right)^{n+1}+\frac{1}{2}\cdot\left(\frac{2}{5}\right)^{n+1}}{\frac{1}{2}\cdot\left(\frac{4}{5}\right)^{n}+\frac{1}{2}\cdot\left(\frac{2}{5}\right)^{n}}$$

$$= \frac{1}{5}\cdot\frac{2^{n+2}+2}{2^n+1}$$

$$\lim_{n\to\infty} P(B_{n+1} \mid B_1B_2\cdots B_n) = \lim_{n\to\infty}\frac{1}{5}\left(4-\frac{2}{2^n+1}\right)=\frac{4}{5}$$

一直摸出黑球使得 U_1 的概率 $\to 1$

<5>

$$P(U_1 \mid B_1B_2\cdots B_n)$$

$$= \frac{P(U_1B_1\cdots B_n)}{P(B_1\cdots B_n)}$$

$$= \frac{\frac{1}{2}\cdot\left(\frac{4}{5}\right)^{n}}{\frac{1}{2}\cdot\left(\frac{4}{5}\right)^{n}+\frac{1}{2}\cdot\left(\frac{2}{5}\right)^{n}}$$

$$= 1-\frac{1}{2^n+1} \to 1$$

解释同上.

T14. <1> 记 A 队获胜概率为 p.

取 x 元与甲对赌

$$\Rightarrow E = px\cdot\frac{25}{5} - (1-p)\cdot x\cdot 0 + (1-p)(100-x)\cdot\frac{25}{10} - p(100-x)\cdot 0$$

$$\geqslant 100$$

又 $p\in[0,1]$.

$\Rightarrow$ $20\leq x\leq 60$ 此时无论 p 为多少 期望总大于 100 元.

<2>

$$P_{甲}(B) = \frac{5}{20+5} = \frac{1}{5}$$

$$P_{乙}(A) = \frac{10}{15+10} = \frac{2}{5}.$$

$$P_{甲}(B) + P_{乙}(A) = \frac{3}{5} < 1$$

甲乙认为对方队伍的胜率和不超过 1.

这表明必存在一种对赌方案赚钱.

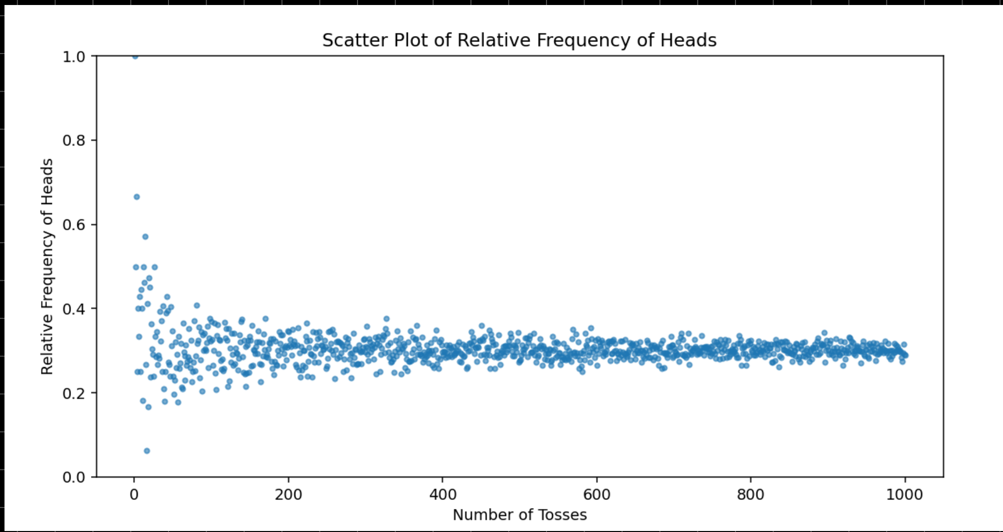
?

T15

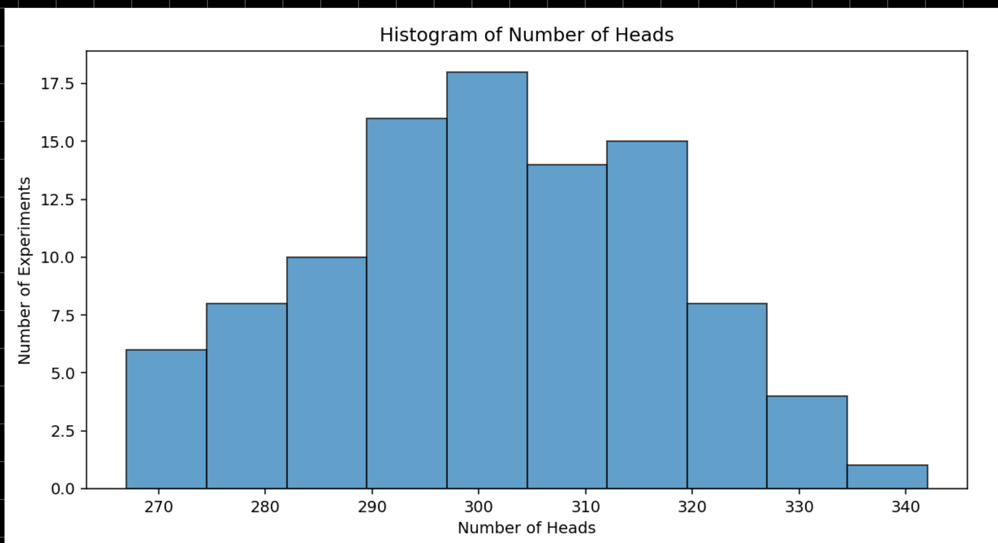
代码如下.

```python
import numpy as np
import matplotlib.pyplot as plt

p = 0.4
n = 3000
repetitions = 100
relative_frequencies = []

for i in range(1, n + 1):
    tosses = np.random.binomial(1, p, i)
    relative_frequency = np.sum(tosses) / i
    relative_frequencies.append(relative_frequency)

plt.figure(figsize=(10, 5))
plt.scatter(range(1, n + 1), relative_frequencies, alpha=0.6, s=10)
plt.xlabel('Number of Tosses')
plt.ylabel('Relative Frequency of Heads')
plt.title('Scatter Plot of Relative Frequency of Heads')
plt.ylim(0, 1)
plt.show()

heads_counts = []

for _ in range(repetitions):
    tosses = np.random.binomial(1, p, n)
    heads_count = np.sum(tosses)
    heads_counts.append(heads_count)

plt.figure(figsize=(10, 5))
plt.hist(heads_counts, bins=10, edgecolor='black', alpha=0.7)
plt.xlabel('Number of Heads')
plt.ylabel('Number of Experiments')
plt.title('Histogram of Number of Heads')
plt.show()

average_heads = np.mean(heads_counts)
expected_heads = p * n

print(f"Average number of heads in 100 experiments: {average_heads}")
print(f"Theoretical expected value: {expected_heads}")
```

<1>



<2>

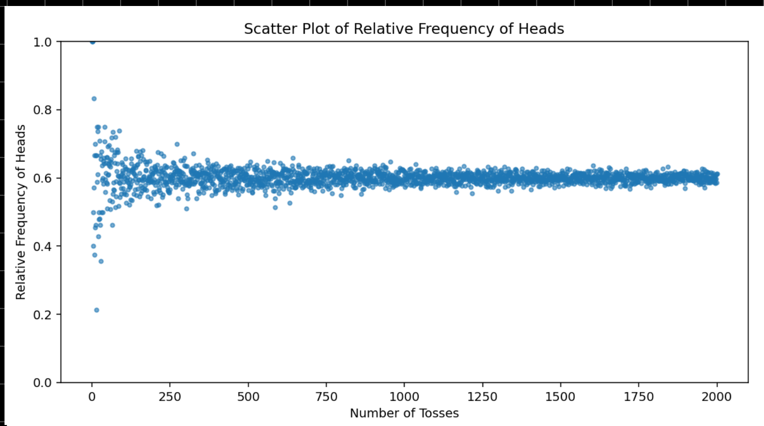


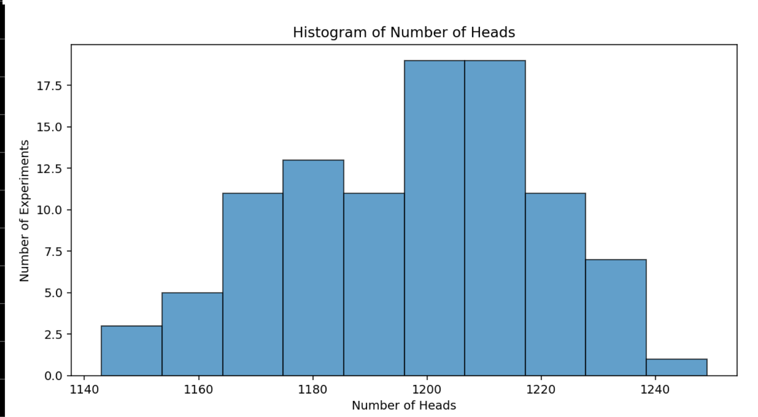
<3>

均值为 300.25

np 为 300

<4> $p=0.6$ $n=2000$ 均值 1200.21 $np=1200$





$p=0.4$ $n=3000$ 均值 1202.45 $np=1200$